

PHILADELPHIA HARBOR, PA, USA

WMO: 997286

Lat: 39.9330N

Lon: 75.1420W

Elev: 66

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.0	(c) 19.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.1	(c) 95.6	(d) N/A	(e) 93.0	(f) N/A	(g) 90.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) 10.2	(f) 100.5	(g) 4.8	(h) 2.8	(i) 6.7	(j) 102.5	(k) 3.9	(l) 104.1	(m) 1.2	(n) 105.7	(o) -2.3	(p) 107.7		
			DB		WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.9	34.5	36.1	44.1	56.4	66.6	75.6	81.0	78.6	71.5	60.1	48.2	40.7
	DBStd	17.80	9.45	8.79	8.64	8.50	7.51	5.99	4.60	4.81	5.91	7.48	7.47	8.18
	HDD50	1575	486	399	224	30	0	0	0	0	0	9	119	308
	HDD65	4219	946	810	649	285	71	4	0	0	10	188	504	752
	CDD50	4460	5	8	41	221	516	768	960	887	645	323	66	20
	CDD65	1630	0	0	1	26	121	322	495	422	206	37	0	0
	CDH74	14956	0	1	10	282	1048	2912	5312	3844	1358	188	1	0
	CDH80	6154	0	0	0	87	360	1199	2505	1546	420	37	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	42.8	3.1	2.8	3.7	3.5	3.5	3.9	4.3	4.0	4.0	3.0	3.1	3.6
	PrecMax	64.6	8.9	6.5	7.3	9.1	7.0	10.6	13.2	19.3	13.1	8.7	9.1	8.9
	PrecMin	29.3	0.5	0.6	0.7	0.5	0.5	0.2	0.7	0.5	0.2	0.1	0.3	0.8
	PrecStd	7.8	1.5	1.3	1.7	1.7	1.6	2.3	2.4	2.9	2.6	1.6	2.0	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	66.4	75.2	88.4	92.6	97.7	100.4	97.4	92.9	84.0	70.9	67.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	58.2	59.2	68.2	81.7	87.9	93.5	96.5	93.6	89.1	79.1	66.9	62.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	53.2	53.1	62.1	76.3	84.0	90.1	93.8	90.7	84.9	75.0	62.6	58.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	48.4	48.3	57.5	71.1	80.2	86.9	91.0	88.0	81.5	71.3	59.5	53.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	13.2	15.4	17.7	18.0	17.2	17.1	15.6	14.7	14.0	12.9	11.5
		MCDBR	20.2	20.3	23.7	25.8	22.9	21.0	20.3	18.8	19.1	18.4	17.9	16.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.315	0.322	0.364	0.395	0.442	0.504	0.508	0.500	0.413	0.362	0.329	0.311
	taud		2.450	2.421	2.365	2.304	2.205	2.073	2.077	2.085	2.302	2.451	2.507	2.513
	Ebn at Noon		270	284	282	279	267	250	247	245	262	266	263	263
	Edh at Noon		25	30	35	40	45	52	51	49	37	29	24	22
(44) All-Sky Solar Radiation	RadAvg		612	899	1188	1537	1712	1875	1908	1663	1375	983	698	532
	RadStd		53	99	95	132	204	151	90	106	150	102	62	54

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218

Nomenclature: See separate page